
CS 6304 PA1

(“Beyond IID: Inductive Biases, Unsupervised Domain Adaptation, Domain Generalization, and Open-Set Recognition”)

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Abstract

Models that match training and test distributions closely (the standard IID, closed-set assumption) can still fail once that assumption is relaxed. This report studies four such failures, all run with one fixed seed (6304): (1) which visual cues a ResNet-50, a ViT-B/16, and CLIP ViT-B/32 rely on under color, cue-conflict, translation, and patch-shuffle interventions on Oxford-IIIT Pets; (2) whether unlabeled target access helps a ResNet-18 adapt from three PACS source domains (Photo, Art Painting, Cartoon) to Sketch via DAN, DANN, and CDAN; (3) whether the same backbone can generalize to Sketch *without* ever seeing it, using ERM, source-only MMD alignment (DAN-DG), and Sharpness-Aware Minimization; and (4) whether a CIFAR-10 classifier can tell when a CIFAR-100 image belongs to none of its ten classes, comparing four post-hoc novelty scores (MSP, MLS, Energy, Mahalanobis) and three trained models (Vanilla, GCSC, PROSER). Four angles, one conclusion: architecture alone does not decide which cues a model uses or how it fails under shift. Alignment that lowers a domain-separability proxy does not automatically improve recognition, and can wipe out class information instead. Local flatness and target-free source invariance do not reliably predict generalization to an unseen domain. And closed-set accuracy is only loosely coupled to a model’s ability to reject genuinely novel inputs. Each claim is backed by controlled comparisons, training-curve diagnostics, and concrete failure cases, not aggregate numbers alone.

Code and machine-readable results: <https://github.com/laeebahafeez/ATML-PA1>

1 Introduction

Ordinary supervised learning assumes training and test data share a distribution, and that every test input belongs to a known class. Both assumptions break in deployment: a model trained on studio photos meets a hand-drawn sketch, and a ten-class classifier meets an eleventh class it has never seen. This report treats those breakdowns as four experiments unified by one question: what should a robust representation preserve, what should it become invariant to, and when should it reject an input rather than force a prediction?

Task 1 asks the preservation question at the level of visual cues. Under interventions that remove color, swap shape and texture, translate the object, or scramble spatial layout, which cues does a CNN, a ViT, and a vision-language model actually use, and does that evidence stay visible in their representations even when predictions do not change? Tasks 2 and 3 ask the invariance question at the level of domains. Given the same three PACS sources and the same unseen Sketch target, does suppressing domain identity in the representation, with unlabeled target access in Task 2 or without it in Task 3, improve recognition, or does it throw away class-discriminative information along with domain identity? Task 4 asks the rejection question: can a classifier restricted to ten known classes tell an ordinary mistake apart from an input that never belonged to any of them, and does that ability track ordinary accuracy or pull apart from it? Section 3 works through each task’s evidence around its research questions; Section 4 returns to the unifying question using evidence gathered across all four.

2 Experimental Setup

Across all four tasks we fix a single seed (6304) for every stochastic step, including splits, initialization, shuffling, augmentation, and random selection, so every reported number traces to a saved configuration or result file.

Task 1. We evaluate on Oxford-IIIT Pets (37 classes; stratified 20% validation split; class-balanced 500-image test subset) rather than the coarser STL-10, since its fine-grained, breed-level labels should make color and texture reliance easier to see in clean accuracy and the size of intervention-induced drops. Three frozen backbones are compared: ResNet-50 [5] (ImageNet1K-V2 weights), ViT-B/16 [1] (ImageNet1K-V1), and CLIP ViT-B/32 [13] (OpenAI-pretrained), each with one linear head trained on top of frozen features (AdamW, lr 10^{-3} , weight decay 10^{-4} , up to 50 epochs with early stopping after 5 epochs without validation-accuracy improvement). CLIP is also evaluated zero-shot (“a photo of a {class}.”). Grayscale is the required color intervention; on top of it we add a fixed 90° hue rotation, on the hypothesis that breed identity is partly color-coded, so a plausible-but-wrong hue should hurt more than removing color entirely (confirmed in 3.1). For translation (0/8/16/32px, four directions) and patch-shuffle (4×4 grid) we report accuracy alongside prediction consistency, $\frac{1}{N} \sum_i \mathbf{1}[\hat{y}(x_i) = \hat{y}(T(x_i))]$, the fraction of predictions unchanged by an intervention. Cue conflicts use full-strength ($\alpha = 1$) AdaIN style transfer [7] across five class pairs in both directions; maximal strength should give the least ambiguous conflicts at the cost of lower coverage, and 240 of 247 attempts passed a rejection rule fixed before any model saw them. We report shape bias, $N_{\text{shape}} / (N_{\text{shape}} + N_{\text{texture}}) \times 100$, alongside coverage, $(N_{\text{shape}} + N_{\text{texture}}) / N_{\text{total}} \times 100$, since shape bias alone hides predictions that fell into neither intended class. Representations are visualized with t-SNE (perplexity 30) rather than UMAP, since its emphasis on local neighbor structure should best show whether transformed examples stay near their clean counterparts; this is quantified as representation stability, $I_T = \frac{1}{N} \sum_i \cos(f(x_i), f(T(x_i)))$, the average cosine similarity between each image’s clean and transformed feature vector.

Tasks 2 and 3. Both tasks share the PACS [9] protocol: Photo, Art Painting, and Cartoon as sources, Sketch as the held-out target, using an ImageNet1K-V1-pretrained ResNet-18 backbone with BatchNorm statistics frozen for every method, so later domain-separability comparisons stay fair. Images are resized to 256px and randomly cropped to 224×224 with horizontal flipping for training (center-cropped for evaluation); every method uses AdamW (lr 10^{-4} , weight decay 10^{-4}) for up to 30 epochs, with early stopping after 5 epochs without improvement on source-validation macro-F1. Each training batch draws 8 examples from each of the 3 source domains (24 total). DAN, DANN, and CDAN additionally draw 24 unlabeled target examples, giving equal source/target batch sizes. Task 2 compares Source-only ERM against DAN [11] (a non-adversarial MMD penalty pulling source and target features together, main $\lambda_{\text{mmd}}=1$), DANN [3] (an adversarial domain discriminator trained via gradient reversal), and CDAN [12] (the same discriminator conditioned on the predicted class); all four clip gradients to a global norm of 1.0 per step (immaterial for Source-only/DAN, binding only once DANN/CDAN’s gradients spike), applied uniformly to keep the optimizer fixed across methods. Task 3 reuses the Task 2 Source-only checkpoint as its ERM baseline, since it never saw Sketch to begin with, and compares DAN-DG (the identical MMD mechanism, main $\lambda_{\text{dg}}=1$, applied only across the three sources) against SAM [2] (main $\rho=0.05$, same optimizer as ERM, no domain-alignment term). We track domain separability throughout (a simple classifier’s held-out accuracy at telling domains apart from frozen features; chance is 1/2 in Task 2, 1/3 in Task 3), and in Task 3 also a local sharpness proxy (loss increase after one small adversarial parameter step).

Task 4. All ten CIFAR-10 classes are known, and unknowns come only from fixed CIFAR-100 classes: a near group (bus, pickup_truck, motorcycle, tractor, wolf, fox, leopard, camel) and a far group (bottle, bowl, chair, clock, keyboard, mushroom, sunflower, wardrobe), 800 images each, never used for training, selection, or calibration. A CIFAR-appropriate ResNet-18 is trained from scratch (random init, not ImageNet-pretrained, with its stem adapted for 32×32 inputs) for two closed-set models, Vanilla and GCSC, using SGD (lr 0.1, momentum 0.9, weight decay 5×10^{-4} , cosine schedule, 100 epochs) with random-crop and horizontal-flip augmentation. GCSC’s only change is inserting RandAugment after crop/flip, which lets us test whether a stronger closed-set classifier is incidentally a better open-set one. For both models we report closed-set accuracy (CSA, ordinary top-1 accuracy on the known classes) alongside near/far AUROC, the threshold-independent area under the ROC curve for separating known from unknown. On Vanilla’s frozen logits and features we compute four post-hoc scores: MSP [6] ($1 - \max_k p_k$), MLS [14] ($-\max_k z_k$), Energy

[10] ($-\log \sum_k e^{z_k}$), and Mahalanobis distance [8] to the nearest class mean. We then fine-tune the selected Vanilla checkpoint into PROSER [16], adding five dummy classifiers trained with a classifier-placeholder loss and a manifold-mixup [15] data-placeholder loss built only from pairs of known CIFAR-10 examples, so no CIFAR-100 leakage occurs. Rejection thresholds are calibrated as the 95th percentile of each score on CIFAR-10 validation data; we report both this rate and the threshold-independent AUROC.

3 Results

3.1 Task 1: Inductive Biases of ResNet, ViT, and CLIP

Table 1: Clean / grayscale / additional-color (hue rotation) / patch-shuffle top-1 accuracy. Source: `step1_clean_baseline.csv`, `step2_color_bias.csv`, `step5_patch_shuffle.csv`.

Model	Clean acc.	Grayscale acc.	Hue-rotation acc.	Patch-shuffle acc.
ResNet-50 (head)	0.908	0.856	0.806	0.784
ViT-B/16 (head)	0.918	0.860	0.844	0.852
CLIP ViT-B/32 (head)	0.788	0.690	0.620	0.504
CLIP ViT-B/32 (zero-shot)	0.756	0.690	0.628	0.532

Table 2: Shape/texture/other decisions (of 240 cue-conflict examples), shape bias, coverage. Source: `step3_shape_bias.csv`.

Model	Shape dec.	Texture dec.	Other dec.	Shape bias (%)	Coverage (%)
ResNet-50 (head)	22	27	191	44.9	20.4
ViT-B/16 (head)	2	1	237	66.7	1.25
CLIP ViT-B/32 (head)	4	8	228	33.3	5.00
CLIP ViT-B/32 (zero-shot)	12	21	207	36.4	13.75

RQ1 (color/cue-conflict). Every model loses more accuracy from hue rotation than from grayscale (Table 1), even though rotation is not geometrically worse; it merely replaces color with a different, consistent color. That pattern supports color being diagnostic for breed identity rather than merely decorative. Cue-conflict coverage is low everywhere (1.25–20.4%), so most stylized images are classified as neither the content nor the style breed. Among decisive predictions every model leans texture or style (ResNet-50: 44.9% shape bias, i.e. 55.1% texture, matching the CNN texture bias reported by Geirhos et al. [4]; both CLIP variants sit at 33–36%). ViT-B/16’s 66.7% figure comes from only 3 decisions out of 240 and should not be read as “ViT is shape-biased.” Its coverage collapse is the more informative fact here, discussed under RQ3.

Table 3: Selected cue-conflict outcomes across the four models. Source: `cue_conflict_predictions.csv`.

Content / style	ResNet-50	ViT-B/16	CLIP (head)	CLIP (zero-shot)	Outcome
Sphynx / Russian Blue	Sphynx (shape)	Persian (other)	Sphynx (shape)	Sphynx (shape)	3/4 agree: shape
Bombay / Min. Pinscher	Bombay (shape)	Birman (other)	Brit. Shorthair (other)	Bombay (shape)	split: 2 shape, 2 other
Russian Blue / Sphynx	Bombay (other)	Abyssinian (other)	Brit. Shorthair (other)	Birman (other)	4/4 other, all different
Beagle / Staffs. Bull Terrier	Sphynx (other)	Bengal (other)	Sphynx (other)	Sphynx (other)	3/4 agree “other”, cross-species
Yorkshire / Scottish Terrier	Bombay (other)	Persian (other)	Brit. Shorthair (other)	Bombay (other)	4/4 other, all cat breeds

Cue-conflict agreement across models. No image gets a unanimous verdict. The strongest cross-model agreement on a decisive answer is 3-of-4, and that happens on only 4 of 240 images (1 shape, row 1; 3 texture). Agreement instead concentrates on failure: at least 3 models say “other” on 227 of 240 images (94.6%), and all four do so, naming four different breeds as row 3 shows, on 160 of them (66.7%). Row 2’s split is not architecture-clean: ResNet-50 and CLIP zero-shot agree on the correct shape (Bombay), while ViT-B/16 and CLIP’s linear probe land on unrelated breeds instead. So most “other” predictions look like genuine failures to resolve either identity, not muted votes for the unseen cue. Rows 4 and 5 show why this happens, not merely that it does: both content and style classes there are *dog* breeds, yet three or four of the four models place the blended image in a *cat* breed instead, never proposing any dog at all (Section 4 discusses the mechanism). That is the clearest evidence that most “other” verdicts reflect the intervention producing ambiguous or misleading evidence, rather than the models being generically cautious.

RQ2 (translation/patch-shuffle). In Figure 1, accuracy stays nearly flat for ResNet-50 and ViT-B/16 out to 32px, while CLIP sits roughly 10 points below its own clean baseline at every displacement including 0, a calibration gap rather than a translation effect, since it does not widen with distance. Consistency tells a different story. ResNet-50 and ViT-B/16 stay at 0.95–0.99, but CLIP drops to 0.86–0.90: roughly 1 in 8 CLIP predictions flip under a shift that barely moves its aggregate accuracy, plausibly because CLIP’s 32×32 patches make a 32px shift exactly one patch width, the worst possible grid misalignment. Patch-shuffling hurts CLIP far more (head: -28.4 pts, consistency 0.51) than ResNet-50 (-12.4) or, notably, ViT-B/16 (-6.6). The supervised ViT is, in fact, *least* disrupted by destroying global layout, which cuts against the intuition that positional tokens should make it the most layout-sensitive model. Pretraining objective looks like the better explanation than architecture here: CLIP’s contrastive training rewards holistic composition, so scrambling patches destroys exactly the evidence it relies on.

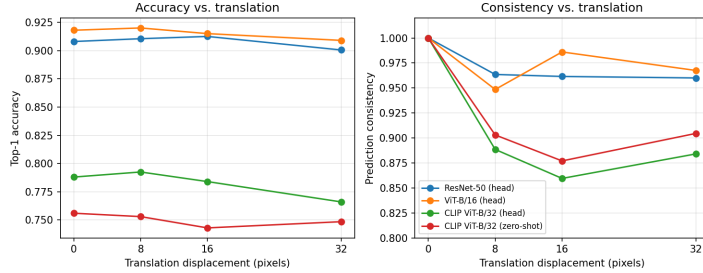


Figure 1: Accuracy (left) and prediction consistency (right) vs. translation displacement. Source: step4_translation_*.csv.

RQ3 (predictions vs. representations). Table 4 and Figure 2 give two clear mismatches. Under patch-shuffle, CLIP has the worst prediction consistency (0.51) yet the *highest* feature-cosine stability of the three (0.777). Its embedding barely rotates even as its argmax flips on half of all images, because cosine stability only checks overall direction, not which side of the decision boundary a feature lands on. Under cue-conflict, CLIP’s stability (0.591) is an order of magnitude above ResNet-50’s (0.069) or ViT-B/16’s (-0.009 , indistinguishable from orthogonal). Figure 2 shows why: for ViT-B/16, every transformed image, regardless of class, lands in one shared region disjoint from the clean clusters, so style transfer erases class identity entirely, matching its near-zero coverage. CLIP’s transformed images also separate from its clean clusters but keep enough per-image structure that paired cosine similarity stays high, plausibly reflecting its more stylistically diverse pretraining data. CLIP’s zero-shot classifier is also more robust than its own trained head on every intervention (Table 1) and reaches higher cue-conflict coverage (13.75% vs. 5.00%), which suggests the head, trained on clean images only, overfits a narrower slice of the embedding space than the zero-shot rule uses.

Table 4: Cosine representation stability I_T (1.0 = identical direction). Source: step6_representation_stability.csv.

Model	Grayscale	Patch-shuffle	Translation (32px)	Cue-conflict
ResNet-50	0.855	0.703	0.950	0.069
ViT-B/16	0.794	0.653	0.965	-0.009
CLIP ViT-B/32	0.908	0.777	0.970	0.591

RQ4 (architecture vs. data/supervision). CLIP and ViT-B/16 share a patch-transformer architecture yet respond oppositely to patch-shuffling (-28.4 vs. -6.6 pts), which tracks pretraining objective rather than architecture. CLIP’s much larger cue-conflict stability plausibly reflects its stylistically diverse pretraining data: ViT-B/16, architecturally closer to CLIP than ResNet-50 is, shows the opposite pattern, the lowest stability of the three. One artifact worth separating from representation quality: CLIP’s linear head has mean max-softmax confidence of only 0.064 despite 78.8% accuracy, far below every other model’s 0.77–0.90. That gap comes from an unscaled linear layer on unit-normalized 512-d embeddings producing small logits, unlike CLIP’s own zero-shot rule, which applies a learned $\approx 100\times$ logit scale to the same embeddings. It is a calibration artifact of how the head was fit, not evidence about the CLIP representation itself.

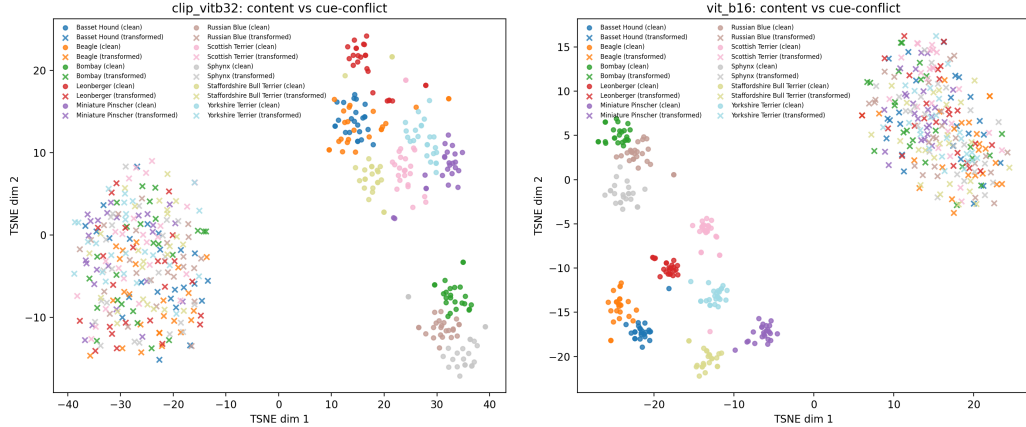


Figure 2: t-SNE of clean vs. cue-conflict features, CLIP ViT-B/32 (left) and ViT-B/16 (right). Circles = clean, crosses = transformed, color = ground-truth class.

3.2 Task 2: Domain Adaptation (PACS, Sketch as target)

Table 5: Per-source-domain validation performance and mean-source performance. Source: task2/results/main_comparison.csv.

Method	Photo		Art		Cartoon		Mean src.	
	Acc.	F1	Acc.	F1	Acc.	F1	Acc.	F1
Source-only	0.9760	0.9718	0.9220	0.9203	0.9254	0.9304	0.9411	0.9408
DAN	0.9670	0.9600	0.9073	0.9064	0.9296	0.9217	0.9346	0.9294
DANN	0.9189	0.9039	0.8512	0.8476	0.8443	0.8315	0.8715	0.8610
CDAN	0.9700	0.9649	0.9122	0.9099	0.9318	0.9405	0.9380	0.9384

Table 6: Target (Sketch) performance, change relative to Source-only, and domain separability. Source: task2/results/main_comparison.csv.

Method	Target acc.	Target F1	Δ tgt. acc.	Domain sep.
Source-only	0.6976	0.6842	0.0000	0.9918
DAN	0.7653	0.7181	+0.0677	0.8228
DANN	0.1117	0.0694	-0.5859	0.9986
CDAN	0.1138	0.1312	-0.5839	0.9973

RQ1 (source-to-target gap). Source-only ERM loses 24.4 points moving to Sketch (0.9411 $\rightarrow$ 0.6976). Person is the largest single failure, at 15.0% Sketch accuracy, with giraffe a distant second at 42.1%; the remaining five classes all reach at least 71.2%. That is consistent with photographic and cartoon people (and, to a lesser extent, giraffes) looking least like their sketched counterparts, while rigid objects such as house and guitar already transfer well with no adaptation at all.

RQ2 (separability vs. recognition). Table 5 shows source performance is consistent across Photo, Art, and Cartoon for every method, so no single domain is quietly propping up the mean, and the aggregate comparison in Table 6 is not hiding domain-specific behavior. The four methods split sharply. DAN lowers separability from near-ceiling (0.992) to 0.823 and gains 6.8 points on target: separability and recognition move together as intended. That gain concentrates unevenly. DAN’s largest per-class improvements are person (0.15 $\rightarrow$ 0.99, +0.84) and giraffe (0.42 $\rightarrow$ 0.68, +0.26), exactly the two classes Source-only struggled with most, while dog is the one class DAN makes worse (0.71 $\rightarrow$ 0.57, -0.14). This is not a generic accuracy trade-off. DAN’s Sketch dog errors concentrate almost entirely on horse (122 of its misclassified dogs) and person (102), precisely the two classes whose own regions the alignment pulled hardest into agreement with Sketch. That points to the shared feature space it builds not being uniformly better, but reshaping decision boundaries around whichever classes needed the most correction, at some cost to a class like dog that sits geometrically between them. DANN and CDAN instead *collapse*: target accuracy falls 58–59 points to near-random while separability stays at 0.997–0.999, essentially unchanged. The adversarial

objective did not even achieve domain confusion, yet it destroyed almost all class structure anyway. Figure 3 shows why: DANN’s domain loss sits flat near zero for four epochs, then explodes past 20,000 at the checkpoint epoch, while CDAN’s domain and classification losses oscillate together and never converge. Confusions confirm the mechanism: 67 (DANN) and 75 (CDAN) of house’s source-validation examples are predicted person, and together with person’s own jump to 100% this shows both models degenerating into predicting person almost everywhere.

RQ3 (conditional vs. marginal alignment). CDAN’s class-conditioning provides no rescue: its target accuracy (0.1138) is indistinguishable from DANN’s (0.1117), with the same collapse. That makes sense once you consider *why* conditioning is supposed to help. CDAN conditions its discriminator on the classifier’s own predicted class distribution, so the extra signal is only as good as those predictions are. Figure 3 shows CDAN’s classification and domain losses oscillating together rather than one stabilizing the other: once training destabilizes, the predicted-class information CDAN conditions on becomes unreliable itself, leaving conditioning nothing trustworthy to condition on. The decisive factor here is adversarial versus statistical alignment, not marginal versus conditional. DAN’s non-adversarial MMD converges smoothly and helps; both gradient-reversal variants destabilize training and hurt regardless of conditioning.

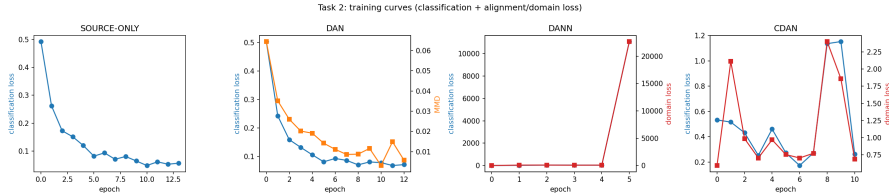


Figure 3: Classification and alignment/domain-loss curves, Source-only/DAN/DANN/CDAN. Source: `task2/results/train_history_*.json`.

RQ4 (controlled alignment strength). Table 7 and Figure 4 sweep $\lambda_{\text{mmd}} \in \{0.1, 1, 10\}$. Target accuracy peaks at the main setting ($\lambda = 1$, 0.765); weak alignment under-corrects (0.716), and strong alignment over-corrects, simultaneously hurting source accuracy (0.943→0.895) *and* raising separability back up (0.880→0.901) instead of continuing to fall. That is optimization instability, not a smooth trade-off. The non-monotonic separability curve in Figure 4 (right) makes this visible directly: separability first drops as λ grows from 0.1 to 1, then rises again at $\lambda = 10$, tracing a V-shape rather than the steadily-decreasing curve a simple “more alignment pressure, less separability” story would predict. At $\lambda = 10$ the MMD term dominates the classification loss enough to destabilize training itself, rather than continuing to smoothly align the two domains. Source accuracy and separability both flag $\lambda = 10$ as unsafe without ever consulting target labels: it is the only setting where source accuracy drops sharply *and* separability climbs back up instead of continuing to fall, the same two-sided warning sign RQ2 used to diagnose DANN and CDAN’s collapse. Comparing only $\lambda = 0.1$ and $\lambda = 1$ on source-side evidence alone would still favor $\lambda = 1$, since it reaches the lower, more-reduced separability (0.823 vs. 0.881) that the alignment objective is designed to produce, while its source-accuracy cost relative to $\lambda = 0.1$ is small (0.935 vs. 0.943) and shows none of $\lambda = 10$ ’s instability. That source-only-justified choice happens to land on the actual best target setting, which suggests the source-side diagnostics are not useless. They simply cannot tell a merely safe setting apart from the optimal one the way target evaluation can.

Table 7: Controlled DAN study: effect of λ_{mmd} . Source: `controlled_study_dan_lambda_mmd.csv`.

λ_{mmd}	Mean src. acc.	Target acc.	Domain sep.
0.1	0.9432	0.7160	0.8805
1.0 (main)	0.9346	0.7653	0.8228
10.0	0.8946	0.6383	0.8901

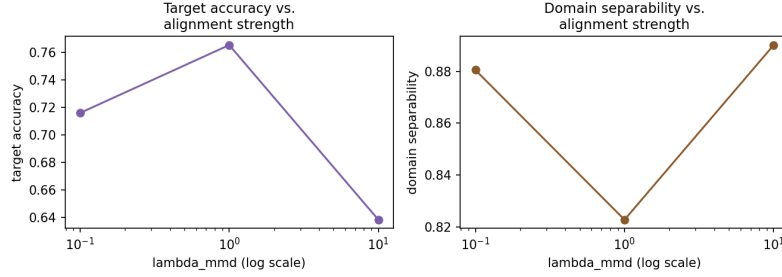


Figure 4: Controlled λ_{mmd} study: target accuracy (left) and domain separability (right) vs. alignment strength. Source: `controlled_study_dan_lambda_mmd.csv`.

3.3 Task 3: Domain Generalization (PACS, Sketch held out)

Table 8: Per-source-domain validation performance, mean- and worst-source. Source: `task3/results/main_comparison.csv`.

Method	Photo		Art		Cartoon		Mean src.		Worst src.	
	Acc.	F1	Acc.	F1	Acc.	F1	Acc.	F1	Acc.	F1
ERM	0.9760	0.9718	0.9220	0.9203	0.9254	0.9304	0.9411	0.9408	0.9220	0.9203
DAN-DG	0.9700	0.9650	0.9098	0.9134	0.9147	0.9231	0.9315	0.9338	0.9098	0.9134
SAM	0.9760	0.9730	0.9439	0.9418	0.9382	0.9447	0.9527	0.9532	0.9382	0.9418

Table 9: Sketch performance, change relative to ERM, source-domain separability, and sharpness. Source: `task3/results/main_comparison.csv`.

Method	Sketch acc.	Sketch F1	Δ vs. ERM	Dom. sep.	Sharpness
ERM	0.6976	0.6842	0.0000	0.8600	0.2788
DAN-DG	0.6808	0.6262	-0.0168	0.7700	0.2797
SAM	0.6455	0.6719	-0.0522	0.8633	0.1209

RQ1 (do source metrics predict Sketch?). Poorly, and in the wrong direction. SAM has the best mean- and worst-source accuracy of the three, yet the *worst* Sketch accuracy (-5.2 pts); DAN-DG has the weakest source numbers but a smaller Sketch drop (-1.7). Table 8 confirms this is not one weak domain distorting the mean: all three sources move together for every method. Per-class evidence shows why the aggregate misleads. DAN-DG’s house collapses on Sketch from 93.75% to 27.5% (confused mostly as giraffe and elephant), the same house that, per Task 2’s RQ1, already transferred well with *no* adaptation at all, so this failure is specific to DAN-DG’s cross-source alignment rather than an inherently hard class. SAM’s dog drops 22.5 points, with 276 of its Sketch confusions landing on elephant alone: large, class-specific failures that are invisible in any aggregate source number.

RQ2 (does DAN-DG’s invariance help?). DAN-DG meets its own target: source-domain separability falls from 0.86 to 0.77. But that invariance does not transfer, since Sketch accuracy ends up slightly *worse* than ERM’s. The house failure above suggests the alignment removed information useful for separating rigid-object silhouettes, not merely domain identity.

RQ3 (does SAM’s flatness predict Sketch performance?). SAM’s sharpness proxy (0.121) is under half of ERM’s (0.279) and DAN-DG’s (0.280, unchanged from ERM as expected). But ranked by flatness, $\text{SAM} \ll \text{ERM} \approx \text{DAN-DG}$, while ranked by Sketch accuracy, $\text{ERM} > \text{DAN-DG} > \text{SAM}$: the exact reverse order. Local stability, by this proxy, does not predict Sketch generalization here, and the proxy’s own definition suggests why not. It perturbs parameters by a small step and measures the loss increase *on held-out source data*, so it only certifies that the loss surface is flat in directions relevant to the three observed sources. Sketch is a large, qualitatively different shift (line drawings versus photos, paintings, and cartoons), not a small perturbation along any direction SAM’s gradient-ascent step would explore. A solution can therefore be flat with respect to source perturbations while still sitting in a region that generalizes poorly along the specific direction separating sources from Sketch.

Flatness here is a claim about the neighborhood SAM actually samples, not about robustness to shift in general.

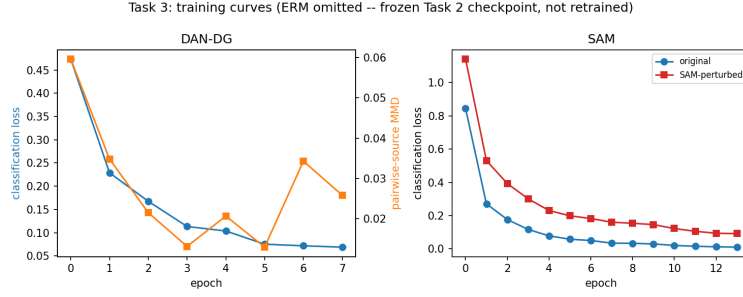


Figure 5: Classification loss (MMD penalty for DAN-DG; original vs. SAM-perturbed loss for SAM). ERM reuses the Task 2 Source-only checkpoint, so has no curve of its own.

The controlled study (Table 10 and Figure 6) sharpens this picture. The *weakest* setting, $\lambda_{dg} = 0.1$, gives the best Sketch accuracy of any Task 3 run (0.727, beating ERM), while the main setting under-performs ERM and $\lambda_{dg} = 10$ collapses training outright (source accuracy 0.381, sharpness 1.90). Separability does not explain the ranking either: $\lambda = 0.1$'s 0.797 exceeds $\lambda = 1$'s 0.770, yet generalizes better. Useful invariance strength has a narrow range, and either too little or too much (or outright optimization failure) can dominate the outcome. Figure 6 (right) shows the sharpness proxy's sudden, order-of-magnitude jump between $\lambda = 1$ and $\lambda = 10$ (log scale, $0.28 \rightarrow 1.90$) lining up exactly with the source-accuracy collapse (left panel) rather than a gradual trade-off, consistent with $\lambda = 10$ destabilizing optimization outright rather than simply over-aligning.

Table 10: Controlled DAN-DG study: effect of λ_{dg} . Source: `controlled_study_dan_dg.csv`.

λ_{dg}	Mean src. acc.	Sketch acc.	Source dom. sep.	Sharpness
0.1	0.9508	0.7272	0.7967	0.2403
1.0 (main)	0.9315	0.6808	0.7700	0.2797
10.0	0.3814	0.1682	0.7000	1.8977

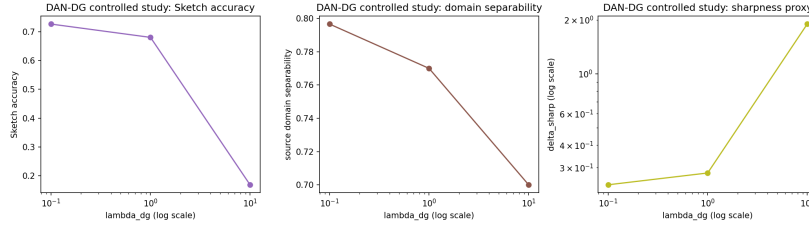


Figure 6: Controlled λ_{dg} study: Sketch accuracy, source-domain separability, and sharpness proxy (log scale) vs. alignment strength. Source: `controlled_study_dan_dg.csv`.

RQ4 (target-aware DAN vs. target-free DAN-DG). Both apply the same MMD mechanism at $\lambda = 1$; the only difference is whether the aligned target is real unlabeled Sketch (DAN) or only the three sources (DAN-DG). DAN reaches 0.765 Sketch accuracy and DAN-DG reaches 0.681, an 8.5-point gap under otherwise matched settings, and DAN-DG's best controlled configuration (0.727) still falls short of DAN's main result. Table 11 makes the mechanism behind that gap concrete at the class level, comparing DAN's Task 2 changes directly with DAN-DG's Task 3 changes for the same seven Sketch classes.

Both mechanisms agree closely on *person* and *giraffe*, the two classes Source-only/ERM struggled with most (+0.84/+0.36 and +0.26/+0.27). Both classes are already somewhat inconsistent in appearance across only Photo, Art, and Cartoon, so pulling those three domains together, which is all DAN-DG ever does, already builds a more style-general notion of “person” or “giraffe” that happens to transfer to Sketch, with no need to have ever seen a Sketch image. The two methods sharply disagree on *house*: DAN leaves it essentially untouched (+0.06), while DAN-DG collapses it by 66 points, confused mostly as *giraffe* and *elephant*. *house* was already one of the best-transferring classes under no adaptation at all (Task 2's RQ1), so it did not need correcting. DAN can tell this, because it gets to see (unlabeled) real Sketch houses during training and can implicitly calibrate

Table 11: Per-class Sketch accuracy change vs. the shared Source-only/ERM baseline: target-aware DAN (Task 2) vs. target-free DAN-DG (Task 3). Source: task2/results/per_class_dan_vs_source_only.csv, task3/results/per_class_dan_dg_vs_erm.csv.

Class	Baseline acc.	DAN Δ (Task 2)	DAN-DG Δ (Task 3)
person	0.150	+0.838	+0.363
giraffe	0.421	+0.260	+0.270
horse	0.721	+0.063	+0.042
guitar	0.924	+0.058	-0.132
elephant	0.845	-0.062	+0.069
dog	0.712	-0.141	-0.361
house	0.938	+0.063	-0.663

how much adjustment each class actually needs. DAN-DG cannot, since it never sees Sketch, so it applies the same cross-source alignment pressure to house as to every other class, distorting a representation that did not need to move until it drifts into unrelated classes’ regions. dog shows the same asymmetry more mildly (-0.14 vs. -0.36): both degrade, but DAN-DG degrades it over twice as much with no target-side signal to limit the damage. Together, this suggests the main value of unlabeled target access is not raising the ceiling on the hardest classes, since generic cross-source alignment already does much of that job blind, but knowing *where alignment should stop*: avoiding damage to classes that were already fine.

This is suggestive, not a clean causal isolation, for three concrete reasons. First, it uses a single seed (6304) throughout, so we cannot rule out that the specific per-class pattern, rather than the general direction of the effect, is partly a property of this one run. Second, DAN’s Sketch access is not a single isolated bit flipped on top of an otherwise identical procedure: every DAN training batch literally contains 24 additional (unlabeled) Sketch examples that DAN-DG’s batches never see, so DAN’s advantage bundles together “having any access to real target structure” with “training on a different, larger effective batch each step,” and we cannot cleanly attribute the gap to the former alone. Third, the two methods were not explored with equal thoroughness. The controlled study (Table 10) shows DAN-DG is quite sensitive to λ_{dg} , with its weakest setting beating its own main result, but DAN’s λ_{mmd} sweep (Table 7) uses the same three-point grid rather than an exhaustive search for its own best configuration. So this compares each method’s *default* recipe, not the best each mechanism could plausibly achieve, and a wider search on either side could narrow or widen the 8.5-point gap reported here.

3.4 Task 4: Open-Set Recognition (CIFAR-10 known / CIFAR-100 unknowns)

Table 12: MSP / MLS / Energy / Mahalanobis on the frozen vanilla model. Source: scores_comparison_vanilla.csv.

Score	Near AUROC	Far AUROC	All AUROC	Near rej.	Far rej.	All rej.
MSP	0.8235	0.8921	0.8578	0.2950	0.4450	0.3700
MLS	0.8070	0.8862	0.8466	0.3288	0.5163	0.4225
Energy	0.8073	0.8867	0.8470	0.3288	0.5263	0.4275
Mahalanobis	0.7984	0.9149	0.8567	0.2825	0.5075	0.3950

Table 13: Vanilla / GCSC / PROSER: CSA and near/far OSR metrics (MLS common score; PROSER’s own placeholder score as an extra row). Source: models_comparison.csv.

Model (score)	CSA	Near AUROC / rej.	Far AUROC / rej.
Vanilla (MLS)	0.9474	0.807 / 0.329	0.886 / 0.516
GCSC (MLS)	0.9504	0.811 / 0.343	0.891 / 0.579
PROSER (MLS)	0.9435	0.804 / 0.301	0.858 / 0.338
PROSER (placeholder score)	0.9435	0.792 / 0.318	0.885 / 0.489

RQ1 (semantic similarity and rejection). Near AUROC (0.80–0.82) trails far AUROC (0.89–0.91) for every score, as designed. Among the ten most confidently-accepted failures in each group (vanilla, MLS), bus→truck recurs most on the near side (6 of 10), followed by pickup_truck→automobile (3 of 10) and a single fox→cat. Two vehicle labels, truck and

automobile, absorb 9 of these 10 near failures between them. On the far side no label dominates nearly as much: mushroom→frog recurs 3 times, bottle→cat and chair→airplane twice each, and bowl splits between cat and ship once each, with one clock→bird. Five different known labels absorb five distinct far classes, each only a handful of times, versus two labels absorbing three near classes almost entirely. Near failures are semantically coherent (bus/truck and pickup_truck/automobile are genuinely similar vehicle shapes; fox/cat share size and posture), matching why these classes were grouped as “near” in the first place and why they collapse onto so few labels. Far failures look more idiosyncratic and spread thinly across more labels: rarer overall, but more like incidental feature overlap than category confusion.

Table 14: Selected near/far failures (vanilla, MLS): most confident incorrect acceptances. Source: failures_near.csv, failures_far.csv.

Type	Unknown class	Predicted class	Score	Threshold
Near	fox	cat	-11.593	-5.764
Near	bus	truck	-11.150	-5.764
Near	bus	truck	-11.021	-5.764
Far	bowl	cat	-12.282	-5.764
Far	bottle	cat	-11.697	-5.764
Far	bowl	ship	-11.329	-5.764

RQ2 (what each score captures). MSP has the highest near AUROC (0.8235); Mahalanobis has the highest far AUROC (0.9149) and the lowest near AUROC (0.7984), the opposite ranking. MSP’s softmax-bounded margin is most sensitive near a real decision boundary but saturates once a prediction is already confident, so it cannot distinguish “fairly sure” from “extremely sure.” MLS keeps the raw, unnormalized top logit instead, so it stays informative in exactly that saturated regime. Energy aggregates *all* class logits via log-sum-exp rather than only the top one, which additionally captures diffuse evidence with no single dominant class; that is why MLS and Energy track each other closely without being identical. Mahalanobis ignores the logits entirely and uses only feature-space distance, so it is most sensitive to gross shift regardless of classifier confidence. Logit magnitude proves most informative not by AUROC but at the actual operating threshold: MLS and Energy give the two highest far-rejection rates of all four scores (51.6%, 52.6%) despite Mahalanobis’s better far AUROC, and similarly reject more near unknowns (32.9%) than MSP (29.5%) despite MSP’s higher near AUROC. Unbounded logit magnitude grows the further an input sits from training data, which can separate outliers well at one specific cutoff even when a bounded or purely geometric score ranks better on average.

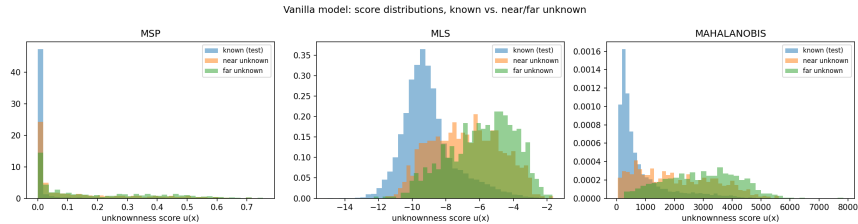


Figure 7: MSP, MLS, and Mahalanobis score distributions, known (test) vs. near vs. far unknown, frozen vanilla model.

RQ3 (does GCSC’s augmentation help?). Both, mildly, and never at a cost. GCSC improves closed-set accuracy (0.9474→0.9504) and both near and far AUROC (+0.4, +0.5 pts). The larger effect shows up at the calibrated operating point for far unknowns: rejection jumps 51.6%→57.9%, against a small 32.9%→34.3% move for near. Stronger closed-set augmentation builds somewhat more compact known-class regions that incidentally flag grossly different inputs better, with no open-set objective involved at all. The near/far asymmetry follows from what “more compact” can and cannot fix. A far unknown sits nowhere near any known-class region to begin with, so shrinking those regions’ boundaries inward directly widens the empty space around it and makes it easier to flag. A near unknown, a wolf for instance, sits close to a genuinely similar known class (dog) in feature space no matter how tightly that known region is drawn. Tightening the boundary does not move the wolf itself further from the dog region, so a near-miss stays a near-miss almost independent of how compact the known classes become.

RQ4 (does PROSER improve rejection?). No, in this run, and not only at the far extreme. PROSER’s CSA (0.9435) is below Vanilla and GCSC, and neither of its scores beats Vanilla’s AUROC or rejection on *either* side. With MLS, near AUROC and rejection are both slightly worse than Vanilla’s (0.807→0.804, 0.329→0.301), and far is where the real damage shows up (AUROC 0.886→0.858, rejection 51.6%→33.8%). With its own placeholder score, far rejection (48.9%) still trails Vanilla, and near AUROC (0.792) trails as well. This is a more complete failure than the placeholder mechanism’s own logic would predict. The classifier- and data-placeholder losses are specifically meant to tighten the boundaries *between pairs of known classes*, which should, if anything, help most with near-misses, yet even the near-unknown numbers did not improve. Its manifold-mixup placeholders interpolate only between pairs of the ten known classes, a plausible proxy for a near-miss between two known classes in principle, but the near-unknowns actually tested here (real CIFAR-100 classes like *fox* or *wolf*) are not literal blends of two CIFAR-10 classes, so even that intended sweet spot was not reliably captured. Interpolation between known classes proves insufficient both for a real, semantically unrelated far-unknown approaching from outside that space altogether, and, more surprisingly, for the near-unknowns it was arguably best positioned to help. This reads as a legitimate negative result rather than a training failure: checkpoint selection followed the same protocol as Vanilla and GCSC.

Positive augmentation vs. PROSER. GCSC (implicit, general augmentation) and PROSER (explicit, targeted placeholder training) are two different strategies for the same trade-off, directly comparable against the same Vanilla baseline with the same MLS score. Despite the more targeted design, the general-purpose strategy wins on every number reported above: GCSC helps closed-set accuracy and all six AUROC/rejection figures; PROSER helps none of them. Section 4 discusses why: what each method’s proxy for “unknown” looks like during training determines what kind of real unknown it generalizes to.

4 Cross-task Discussion

Visual cues and distribution shift. Task 1 shows color is more than decorative for every backbone, and CLIP is uniquely hurt by losing global layout; PACS shift (Tasks 2 and 3) changes color statistics, rendering style, and layout all at once. This is a qualitative parallel, not a causal claim, since the datasets, architectures, and training regimes all differ. But the consistent theme is that no cue category is safe to assume irrelevant, and vulnerability tracks pretraining objective as much as architecture. Two Task 1 results make the “objective, not architecture” half of that claim concrete rather than qualitative. First, CLIP and ViT-B/16 share a patch-transformer architecture yet respond oppositely to patch-shuffling (−28.4 vs. −6.6 accuracy points). CLIP’s contrastive pretraining rewards holistic image-text composition, so scrambling patches destroys exactly the evidence it relies on, while the supervised ViT is comparatively undisturbed despite its positional tokens. Second, the cue-conflict examples where full-strength AdaIN transfer produces images that three or four of four models place in a *cat* breed despite both source classes being dog breeds (Table 3, rows 4–5) show that a nuisance-cue intervention can manufacture inputs that are, in the model’s own feature space, genuinely closer to a class neither source class intended, independent of any single architecture’s inductive bias.

Invariance vs. discriminability. Across Tasks 2 and 3, a lower separability or sharpness number is neither necessary nor sufficient for better recognition. DAN is the one case where lower separability and higher target accuracy move together, with stable training curves. Because MMD is a smooth statistical distance, it provides a well-behaved gradient throughout training, so the model can reduce it gradually while classification loss keeps improving alongside it. DANN and CDAN fail to even reduce separability while collapsing class structure, which the divergent domain-loss curves diagnose directly: their minimax objective pits the discriminator against the feature extractor, and once one side starts to dominate (Section 3.2, RQ2 and RQ3), the resulting gradients stop being informative about domain identity at all and instead simply push features toward whatever destroys the discriminator’s current signal, class structure included. DAN-DG and SAM each hit their own targeted proxy (lower source separability, lower sharpness) without collapse, yet neither beats plain ERM on Sketch. SAM, the best source-side performer on every proxy, generalizes worst, because both proxies are measured only with respect to the three observed sources and have no mechanism to certify anything about a direction as different as Sketch. These proxies need to be checked against class-level and target-domain evidence, not trusted on their own.

The importance of target domain access. With unlabeled target access, DAN reaches 0.7653 Sketch accuracy. The identical mechanism restricted to observed sources, DAN-DG, reaches only 0.6808, below the no-adaptation ERM baseline (0.6976). Section 3.3’s RQ4 breaks this down per class: target access matters less for raising the ceiling on the hardest classes (person and giraffe improve similarly either way) and more for knowing *where alignment should stop*. DAN-DG, blind to Sketch, keeps “correcting” an already-fine house until it breaks, while DAN does not. That section also details why this falls short of a clean causal isolation: one seed, a batch-composition confound, and asymmetric hyperparameter search.

Recognition vs. rejection. GCSC shows a better closed-set classifier can also be a modestly safer one: accuracy and both near and far rejection improve together. PROSER breaks that pattern in both directions. It costs closed-set accuracy *and* fails to improve rejection, showing an explicitly open-set method can still underperform a plain classifier if its proxy unknowns (interpolations between known classes) do not resemble the real unknowns it is tested against. The contrast is informative about *why*, not merely *that*. GCSC’s RandAugment perturbs real known-class images with no explicit notion of “unknown” at all, so any rejection benefit is a side-effect of a generally more compact known-class representation, useful against any kind of shift. PROSER’s manifold-mixup placeholders are explicitly targeted at carving out low-confidence regions, but only *between* pairs of known classes, which can plausibly help a near-miss between two known classes but supplies no signal about the far side of the decision boundary, where most real unknowns, especially far ones, actually sit. A general-purpose regularizer with no open-set objective transferred to rejection better here than a purpose-built method whose proxy unknowns did not resemble the real ones it was tested against. Every method’s near-unknown numbers also move less than its far-unknown numbers, echoing Task 1’s coverage warning: aggregate numbers can hide which regime actually changed.

Overall synthesis. The safest generalization here is a negative one: architecture, a domain-invariance proxy, a flatness proxy, and closed-set accuracy are each informative but individually unreliable stand-ins for the property they are meant to predict, and each needs class-level or target-domain checking. What should be preserved is class-discriminative structure, which DANN and CDAN, and to a lesser extent DAN-DG and PROSER, damaged while pursuing a different, plausible objective. What should become invariant is only nuisance variation that carries no class information. DAN’s moderate, stable alignment is the one case that struck this balance, while every method that pushed invariance too hard destroyed discriminative structure along with it. What should be rejected rather than forced into a label is exactly what Task 4 formalizes, with the near/far distinction setting how hard that judgment is and which score suits it.

5 Conclusion

Four experiments framed as one question point to the same answer from different directions. Clean accuracy, architecture family, and simple invariance or flatness proxies are each informative but individually unreliable guides outside the IID, closed-set setting they were measured under. Task 1 shows architecture does not cleanly predict cue reliance or spatial sensitivity; pretraining objective and data matter at least as much. Tasks 2 and 3 show a method can satisfy its own stated objective without improving, and in the adversarial cases while actively destroying, the recognition it was meant to serve. Task 4 shows closed-set accuracy and open-set rejection are only loosely coupled. Section 4 grounds each of these claims in concrete, class-level evidence rather than aggregate numbers alone. The practical implication is to treat shape bias, domain separability, sharpness, and closed-set accuracy as diagnostics checked against class-level and target-distribution evidence, not as targets optimized and trusted in isolation. When a proxy for a hard-to-measure quantity is unavoidable, ask what kind of failure that proxy is blind to before trusting it.

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